



Automated Plant Disease Detection

Using Deep Learning on Saudi-Relevant Crops

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Abstract: AI for Agriculture

1

Global Impact

Plant diseases threaten food production, especially in Saudi Arabia.

2

MobileNetV2 System

Developed for tomato, pepper, potato disease classification.

3

High Accuracy

Achieved 91% test accuracy across 15 classes.

4

Saudi Vision 2030

Aligns with food security and tech innovation goals.

Introduction: Modernizing Agriculture

Agriculture is vital for food stability and economic growth, a key priority for Saudi Vision 2030.

Traditional disease inspection is slow and subjective.

Deep learning offers accurate, scalable, and automated diagnosis.





Problem Statement: Complexities of Diagnosis

→ **Visual Similarity**

High similarity between different disease types.

→ **Data Challenges**

Dataset imbalance and limited real-field data.

→ **Deployment Needs**

Requirement for lightweight models on mobile devices.

→ **Early Detection**

Crucial to minimize crop loss.

Dataset: Saudi-Relevant Crops

Derived from PlantVillage dataset.

- Tomato
- Pepper (Bell Pepper)
- Potato

15 classes: diseased and healthy leaves.

Total: 68,361 images.



Preprocessing

Resize, normalize, split, encode labels.

Augmentation

Rotation, brightness, zoom, shifting, shear, flips.

Class Balancing

Mitigated underrepresented classes.

Methodology: Deep Learning Workflow

01

Transfer Learning

MobileNetV2 with ImageNet weights, base layers frozen.

02

Custom Head

GlobalAveragePooling2D, Dense layers, Dropout, Softmax.

03

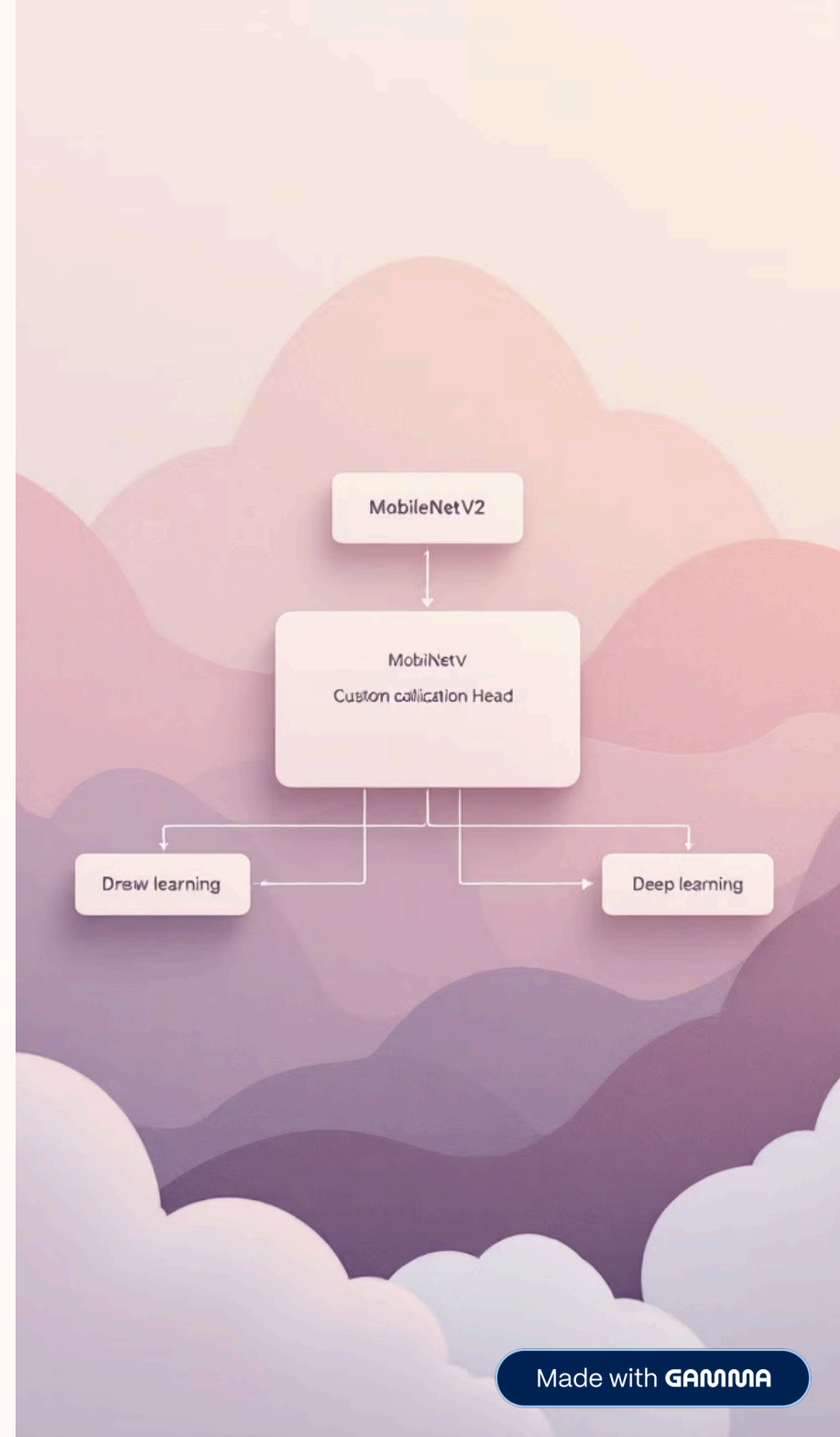
Stage 1 Training

Trained only the custom classification head.

04

Fine-Tuning

Unfroze top convolutional blocks, reduced learning rate.

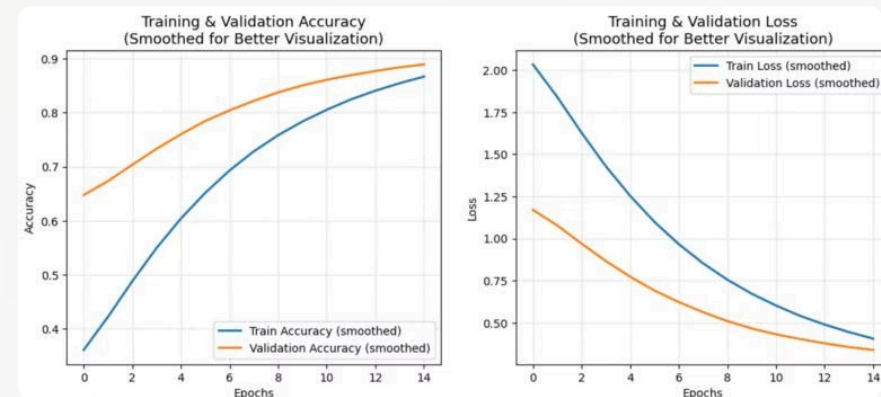


Experiments: Learning Curves

Training and validation accuracy curves show stable, effective learning.

Validation accuracy consistently slightly higher, indicating strong generalization.

Final test accuracy: 91.08%



Results: Overall Accuracy & Confusion

91% Test Accuracy

Strong overall performance across 15 classes.

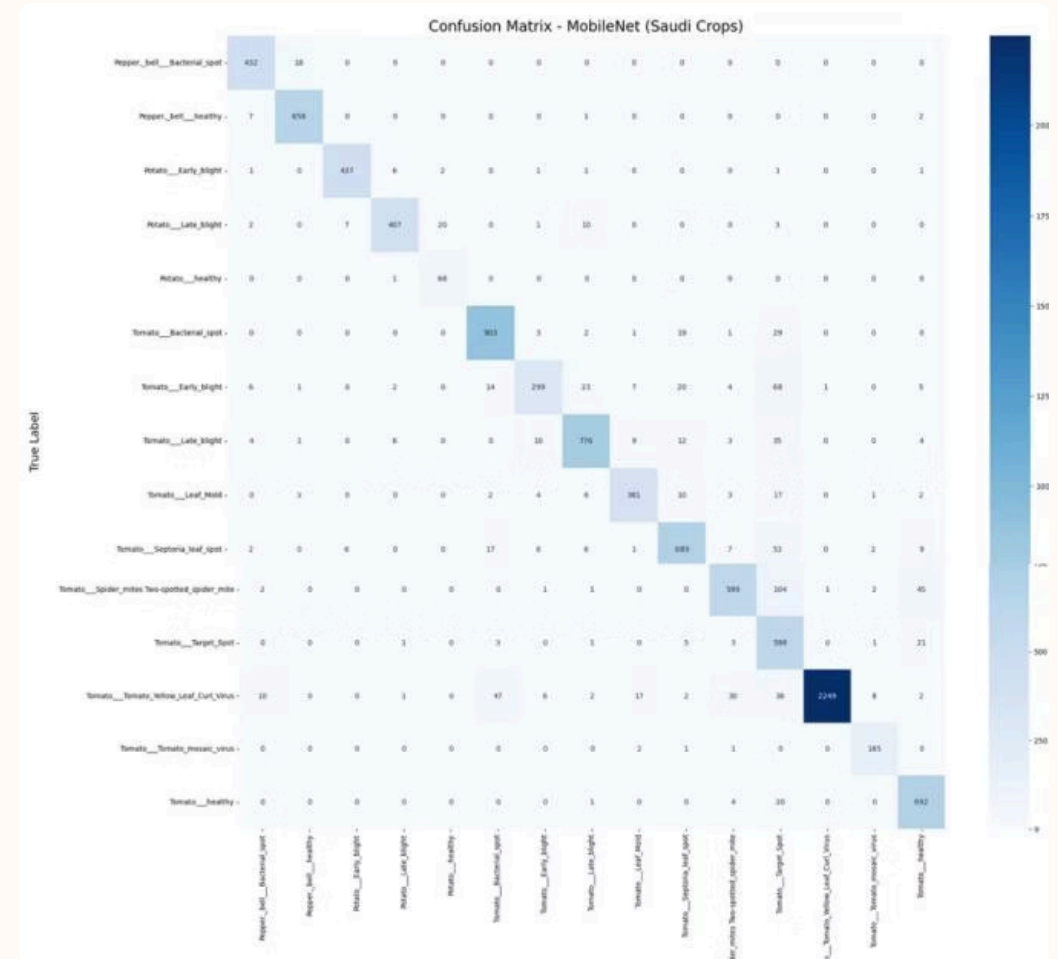
Robust, generalizable features learned.

Confusion Matrix

Strong diagonal dominance.

Some confusion between Early Blight and Late Blight.

Viral classes performed exceptionally well.



Real-World Testing & Misclassification

Most predictions accurate (0.64 to 0.99 confidence).

One healthy potato leaf misclassified as Late Blight due to:

- Domain Shift (lab vs. field images)
- Texture Confusion (natural variations mistaken for disease)
- Lack of Real-Field Training Samples

Future Work & Conclusion

YOLOv8 Detection

Object detection for precise lesion location and multi-disease identification.

Cascaded Hybrid Architecture

MobileNetV2 for coarse detection, EfficientNetB0 for fine-grained classification.

Field Data Expansion

Incorporate real Saudi field images for robustness.

Edge Device Deployment

Mobile apps, drones, IoT sensors for in-field diagnosis.

This project supports Saudi Vision 2030 for smart agriculture and food security.